Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

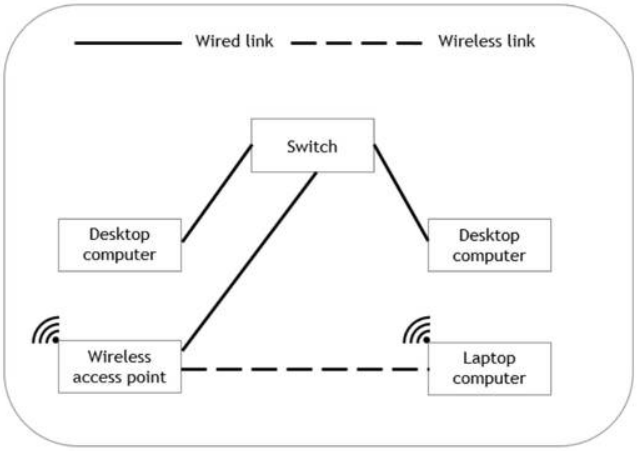
1(a)	Award **one** mark for each correct row: A network of networks **not** limited to a single location: WAN ✓ A network that connects devices within the range of a single user: PAN ✓ A set of hypertext documents accessed through the HTTP protocol: WWW ✓ A system that maps domain names to IP addresses: DNS ✓	Do not award a mark for a row containing more than one tick.	4

Question	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • Transport (layer) (1) • (Data) link / network link / network interface (layer) (1)	Accept • transfer or transmission for Transport Do not accept: • Network layer and Application layer (in the question) • Data access layer, Internet layer (alternative names for network layer) • Data' or 'network' without 'link' or 'interface'	2
1(c)	**The only correct answer is D** *A is not correct because HTTP is used to transfer hypertext* *B is not correct because it is used to view email held on a server* *C is not correct because it is used to download email from a server*		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each correct connection: • One solid line from wireless access point to switch (1) • One solid line from each desktop to switch (1) • One dashed line from laptop computer to wireless access point (1) 	No marks should be awarded if any additional lines have been drawn for each mark point. Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop)	3

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(e)	Award up to **two** marks for a linked description such as: • To find weaknesses in a system (1) fix/repair weakness (1) • To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1) • An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1) • Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1) • To test the system security (1) / sanctioned by the owner (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 1	**12**

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

5(a)	<table><tr><th>P</th><th>Q</th><th>R</th><th>Q OR R</th><th>P AND (Q OR R)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> Award **one** mark for each of: • All possible combinations in columns P, Q and R (1) • Correct values in Q or R (1) • Correct values in final column (1)	Allow follow through for incorrect mark point 2.	3

Question Number	Answer	Additional Guidance	Mark
5(b)	B AND S AND NOT L Award **one** mark for each of: NOT L or B AND S (1) Fully correct (1)	• Any order and ignore brackets	2

Question Number	Answer	Additional Guidance	Mark
5(c)	Any **two** of: • Remove code vulnerabilities in programming languages (1). • Eliminate bad programming practices (1).		2

Question Number	Answer	Additional Guidance	Mark
5(d)	Award **one** mark each for any of: • Copyright (1) • Patent (1) • Licensing (1) • Trademark (1)		3

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

16

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part d.i in this mark scheme extract

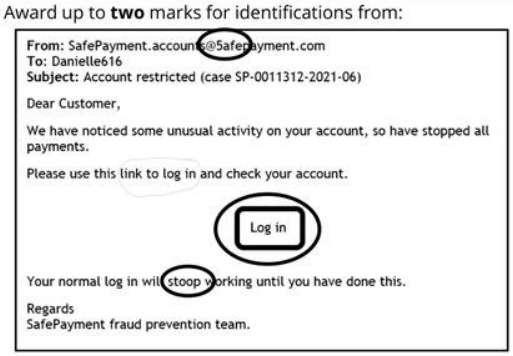
2(a)	**The only correct answer is B** *A is not correct because a password is needed to prevent unauthorised access to the network* *C is not correct because it only provides security after access to the network* *D is not correct because it only provides security after access to the network*		1

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part d.i in this mark scheme extract

2(b)(ii)	Award up to **two** marks for identifications from:  **From:** SafePayment.account@5afepayment.com **To:** Danielle616 **Subject:** Account restricted (case SP-0011312-2021-06) Dear Customer, We have noticed some unusual activity on your account, so have stopped all payments. Please use this link to log in and check your account. Log in Your normal log in will stoop working until you have done this. Regards SafePayment fraud prevention team.	Allow link to log in OR circle around button	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part d.i in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

6(a)(i)	Award up to **two** marks for a linked explanation such as: • Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1) • No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1) • Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1) • Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1) • Easy to find faults (1) all of the tokens will end up on one workstation (1) • Cheap to set up (1) uses minimum cabling (1)	Not a **Comparison** without expansion that fits mark scheme	2
6(a)(ii)	Award up to **two** marks for a linked explanation such as: • It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1) • Can handle high volumes of data traffic (1), because data can travel via multiple routes (1) • Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc • Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark for: • Personal area network / PAN/WPAN		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

6(b)(ii)	Award up to **two** marks for a linked explanation such as: • A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1) • Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award **one** mark from: • A record of activities/specified activity that have taken place on a computer system (1) • Automatic record of what has happened and who did it (1)		1
6(c)(ii)	Award **one** mark from: • To identify suspicious/malicious activity/changes (1) • To increase accountability (1) • To trace a problem back to its source/perpetrator (1) • To find out if any users are using unauthorised applications (capable of putting the network at risk) (1)		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(d)	**Ethical hacking** • Ethical hackers are white hat hackers • Attempt to access the network as a hacker does • Don't attempt to change or steal data • Looking for weaknesses in the network • Weakness pointed out • Weaknesses fixed • Could be employed by the business • Could work for another specialist company • Can include penetration testing **Commercial analysis tools** • Software used to find weaknesses • Can be configured to check for a range of weaknesses • Results/reports generated identifying faults • Weaknesses fixed **Review of network and user policies** • Collection of rules and guidelines that govern the behaviours of network devices/users • Need reviewing because may not comply with new laws and regulations • Reviews should be scheduled		6
		Total for question 6	**15**

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

6(a)(i)	Award up to **two** marks for a linked explanation such as: • Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1) • No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1) • Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1) • Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1) • Easy to find faults (1) all of the tokens will end up on one workstation (1) • Cheap to set up (1) uses minimum cabling (1)	Not a **Comparison** without expansion that fits mark scheme	2
6(a)(ii)	Award up to **two** marks for a linked explanation such as: • It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1) • Can handle high volumes of data traffic (1), because data can travel via multiple routes (1) • Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc • Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark for: • Personal area network / PAN/WPAN		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

6(b)(ii)	Award up to **two** marks for a linked explanation such as: • A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1) • Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award **one** mark from: • A record of activities/specified activity that have taken place on a computer system (1) • Automatic record of what has happened and who did it (1)		1
6(c)(ii)	Award **one** mark from: • To identify suspicious/malicious activity/changes (1) • To increase accountability (1) • To trace a problem back to its source/perpetrator (1) • To find out if any users are using unauthorised applications (capable of putting the network at risk) (1)		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(d)	**Ethical hacking** • Ethical hackers are white hat hackers • Attempt to access the network as a hacker does • Don't attempt to change or steal data • Looking for weaknesses in the network • Weakness pointed out • Weaknesses fixed • Could be employed by the business • Could work for another specialist company • Can include penetration testing **Commercial analysis tools** • Software used to find weaknesses • Can be configured to check for a range of weaknesses • Results/reports generated identifying faults • Weaknesses fixed **Review of network and user policies** • Collection of rules and guidelines that govern the behaviours of network devices/users • Need reviewing because may not comply with new laws and regulations • Reviews should be scheduled		6
		Total for question 6	**15**

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Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

6(a)(i)	Award up to **two** marks for a linked explanation such as: • Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1) • No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1) • Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1) • Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1) • Easy to find faults (1) all of the tokens will end up on one workstation (1) • Cheap to set up (1) uses minimum cabling (1)	Not a **Comparison** without expansion that fits mark scheme	2
6(a)(ii)	Award up to **two** marks for a linked explanation such as: • It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1) • Can handle high volumes of data traffic (1), because data can travel via multiple routes (1) • Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc • Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark for: • Personal area network / PAN/WPAN		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

6(b)(ii)	Award up to **two** marks for a linked explanation such as: • A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1) • Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award **one** mark from: • A record of activities/specified activity that have taken place on a computer system (1) • Automatic record of what has happened and who did it (1)		1
6(c)(ii)	Award **one** mark from: • To identify suspicious/malicious activity/changes (1) • To increase accountability (1) • To trace a problem back to its source/perpetrator (1) • To find out if any users are using unauthorised applications (capable of putting the network at risk) (1)		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(d)	**Ethical hacking** • Ethical hackers are white hat hackers • Attempt to access the network as a hacker does • Don't attempt to change or steal data • Looking for weaknesses in the network • Weakness pointed out • Weaknesses fixed • Could be employed by the business • Could work for another specialist company • Can include penetration testing **Commercial analysis tools** • Software used to find weaknesses • Can be configured to check for a range of weaknesses • Results/reports generated identifying faults • Weaknesses fixed **Review of network and user policies** • Collection of rules and guidelines that govern the behaviours of network devices/users • Need reviewing because may not comply with new laws and regulations • Reviews should be scheduled		6
		Total for question 6	**15**

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 8 of 8 · Question 4 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 8 of 8 · Question 4 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 8 of 8 · Question 4 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark